DA DOME

DECO3500 - JUST 5 VIBES

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WHAT WE HAVE ENHANCED

We're designing for large-scale, multi-stage music festivals to enhance how familiar strangers connect and interact in the gaps between sets. This was shaped through past solutions, user research and three rounds of various lo-fi/mid-fi prototyping.

THE DESIGN JOURNEY

What if we used the overlooked spaces at music festivals to connect familiar strangers while also enhancing the experience itself?

At a multi-stage festival, there's always a stretch of ground between stages.

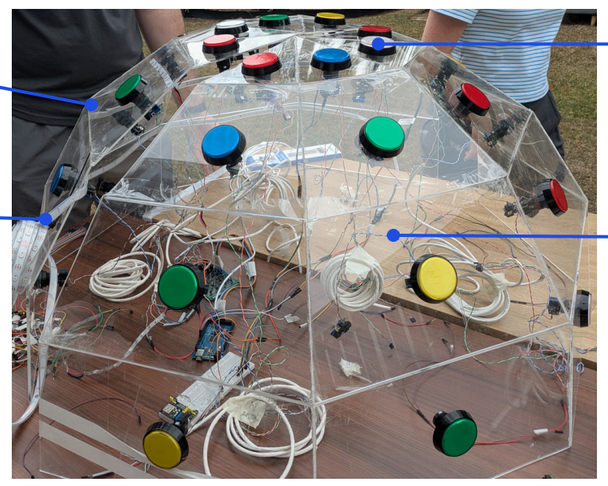
THE EXPERIENCE

***Touch** the buttons. **Listen** to the layered tracks.*
***Build** the song together.*

Da Dome is a geodesic dome-inspired structure covered in buttons. Each button triggers a stem of a song, with bass at the bottom, then drums, keys, and vocals toward the top. The more buttons people press, the richer the track becomes and the brighter the dome glows!

The clear acrylic allow users to see the LED's and mechanics inside

LED light strips light up a specific section when pressed. The lights become more intense when more of the same row is pressed!

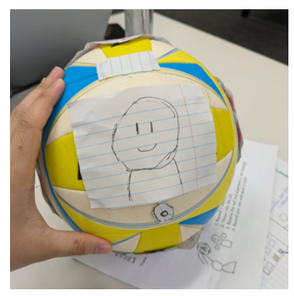


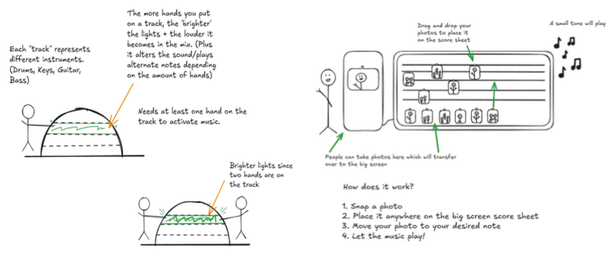
Each row of buttons represents a different instrument

The clear acrylic allow users to see through the dome and be more aware of others' actions

PROTOTYPES

Lo-fi round 1 : Testing the ball and the table separately showed people needed to see or sense a response before they'd commit to interacting, which ruled out the ball's throw-and-hope mechanic and pointed toward something with instant feedback.





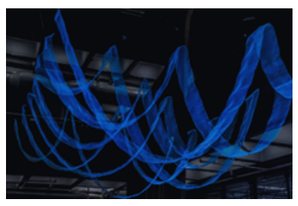
Lo-fi round 2: Pitting *Da Dome* against the photoboard showed that an unfamiliar, game-like object draws people in through curiosity alone, while the photoboard's purpose had to be explained, a strong early signal that the dome's design should stay instruction-free.

Mid-fi prototype: Groups had to point, call out, and compare tracks to work out what each panel did before the song came together, as the cardboard prototype gave them nothing to see, they relied on what they could hear. Design implication: because visual awareness wasn't possible yet, users relied entirely on talking out loud to coordinate, confirming that the final build needed to be see through so the lighting feedback would actually be visible, not hidden.



BACKGROUND RESEARCH

We were inspired by three main previous solutions:



Audience Response Canopy (ARC)
A ribbon display that moved in response to audience energy during a performance.
Design implication: subtle, ambient feedback can affirm a shared crowd mood without demanding direct interaction.



Sounds of Infinity
Public installation that turned sound and movement into light.
Design implication: a low barrier to entry (just making noise) pulled strangers in, watching others try it created a community feel.



REACTable
Tabletop instrument where users link "nodes" to build music together.
Design implication: collaborative, but favours people with music-making skill, a barrier *Da Dome* wanted to avoid.

SOCIAL & MOBILE POSSIBILITIES

Why a Shared Dome?

Da Dome is a shared, physical experience built for music festivals, where everyone is in the same place at the same time. Using the *Designed Experiences for Co-located Interaction* framework, we focused on shared presence and real-time awareness over remote or asynchronous play.

One object, shared control.

Because everyone controls the same dome, people explore the interaction, stay aware of what others are doing, and adjust their behaviour to fit the group.

Button placement encourage collaboration.

Buttons are spaced so that no one person can reach them all. Reach and height differ across a group, so players need to work together by talking and taking turns.

FUTURE DIRECTIONS

Testing with strangers: Our testing so far has mostly used people who already know each other, not the kind of strangers you'd find at a festival. Mixing in people who don't know each other would be a closer match and a better test of whether the dome works as intended.

Testing at a real festival: Live testing at a festival or crowded public space would show us things small group testing can't. One open question from our own notes is whether a stranger joining a group that's already mid-song changes the experience for the people already playing.

USER RESEARCH

Based on our user research, we conducted qualitative interviews with 8 music festival goers to understand the values of our target audience. Our thematic analysis found the following main points:

Low pressure connection: Participants wanted brief ways to connect with strangers nearby, without the pressure of real conversation. As P8 put it, "something that's almost like an icebreaker would definitely make me come out of my comfort zone."

Comfort varies: Comfort with being approached varied a lot between participants, so any interaction needed an easy way to opt in or out.

Unreliable connectivity: Network and phone reliability was the most consistently raised issue across the group. As P4 said, "Often there's no signal, actually... calling, texting just doesn't work." This directly shaped *Da Dome*'s decision to be fully physical, with no app, login, or network dependency at all.

ETHICAL CONSIDERATIONS

No data is collected or stored as all the interactions are physical.

Mid-Fi testing showed us two concerns: groups "hogging" the dome, and people who don't know each other not wanting to join in.

Spacing the buttons stops any one person taking over, but a large group could still crowd others out. *Da Dome* is meant to be a quick stop. One participant told us festival breaks are short and mostly spent eating and resting, so "a lot of that time's already taken away" by the time you reach an activity, making long stays unlikely. If hogging did become a problem, we'd need to test and prototype solutions in future.

Strangers may still feel awkward joining a group that's already using the dome. Because our testing so far hasn't included real festival crowds, we can't yet say how big this problem would be (see Future Considerations).